

Name:

Roll no:

VISWAJYOTHI COLLEGE OF ENGG. & TECH. VAZHAKULAM

Re series

GRAPH THEORY(GT)

MAT206

Time: 1 Hour

Part A

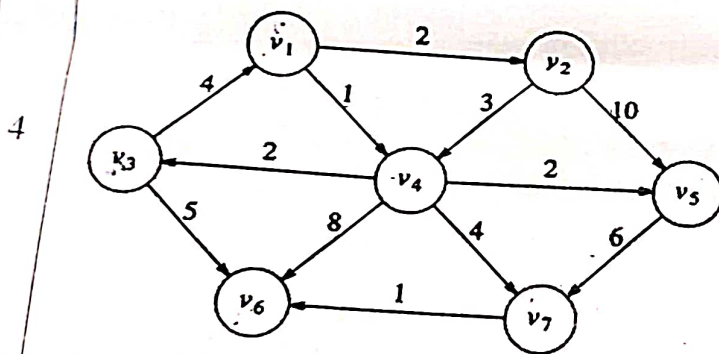
Max: 30 marks

(Answer all questions.)

No	Questions	Marks	CO mapped	Taxonomy
1	Prove that a tree with n vertices has $n-1$ edges.	3	C211.3	Comprehension
2	Draw the dual of complete graph with 4 vertices.	3	C211.4	Comprehension
3	A connected Planar graph has vertices having degree 2,2,2,3,3,3,4,4,5. a) How many edges are there? b) How many faces are there?	3	C211.4	Comprehension

Part B

(Answer any 3 questions)

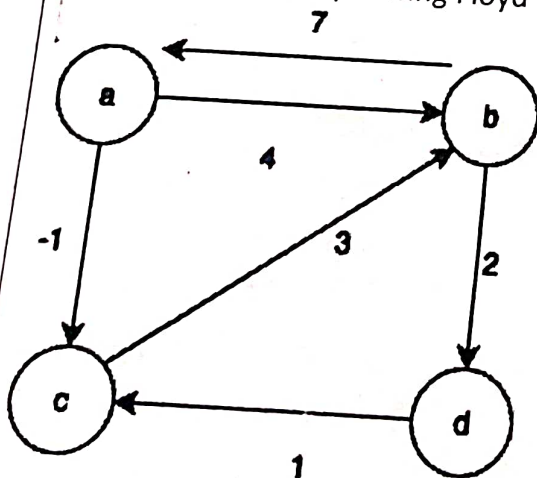
Find a shortest path from vertex V_3 to V_5 Dijkstra's Algorithm

7

C 211.3

Application

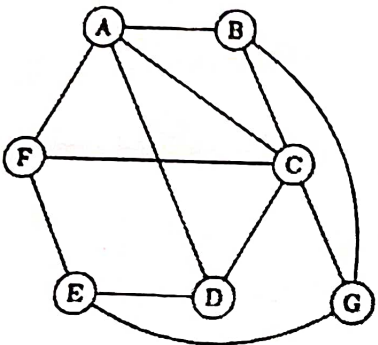
Find the shortest paths between every pair of vertices in the following graph using Floyd Warshall Algorithm



7

C211.3

Application

6	State and prove Euler's formula for Planar Graphs	7	C211.4	Comprehension
7	State Kuratowski's Theorem and use it to prove that the following graph is non planar. 	7	C211.4	Application

Course Coordinator

H. O. D